

Power Systems:

Friction Calculations → Torque Requirement → Motor Selection + Gear ratio

Our initial idea was to use springs to provide the clamping force required for the robot to be in a state of static vertical equilibrium when no power is provided to the motors. The robot must maintain this force even when the increasing downward force by the chains are at maximum. To reduce costs we opted for using 4 rubber bands. We varied the amount of 'turns' and used a Newton-meter to empirically infer the force required to pull the system loaded with our estimated maximum mass of 1.4 kg (estimated mass of the finished robot and the chain at its maximum length). From our observation we concluded that 4 rubber bands, each wrapped twice tethering the body to each leg, had an approximately 5N threshold to initiate upward motion. The 5N threshold was maintained even with varying load scenarios, indicating the rubber bands provided reliable constant resistive force. We only accounted for the upward motion as it is the minimum requirement for the motor.

The torque requirement was calculated as such:

$$T = Fr_{motorised} = ((1.4 \cdot 9.81) + 5.00) \cdot 0.04 = 749.40 \text{ Nmm} = 74.94 \text{ Ncm}$$

With the required torque calculated we began the motor selection process.

Motor Selection Process:

We imported the entire motor stock from RS Components as a .csv file. Firstly sorting it based on price – everything above £50 was excluded. Then sorting it based on torque – excluding motors where our operating torque lies outside the respective motors stall torque-no load speed curve. That left us with 18 contenders from which we settled for the 19.8W 4.5-15V DC Brushed Motor with an output torque of 59Ncm and 143rpm at 15V input. For the main power to the motors we opted for a 12V 2400mAh NiMH rechargeable battery. The motor can take up to 15V but in the specifications, it is rated for 12V so with our current setup it provides 59Ncm of output torque. Furthermore, supplying a 15V motor with 12V acts as a safety clearance from overheating and prolongs motor life. The motor output was insufficient for upward drive, so a reduction gear was added to increase the transported torque to the motorised wheel.

Gear Calculation:

A 13 teeth driving gear and a 39 teeth driven gear give the required torque margin and also retain a small form factor

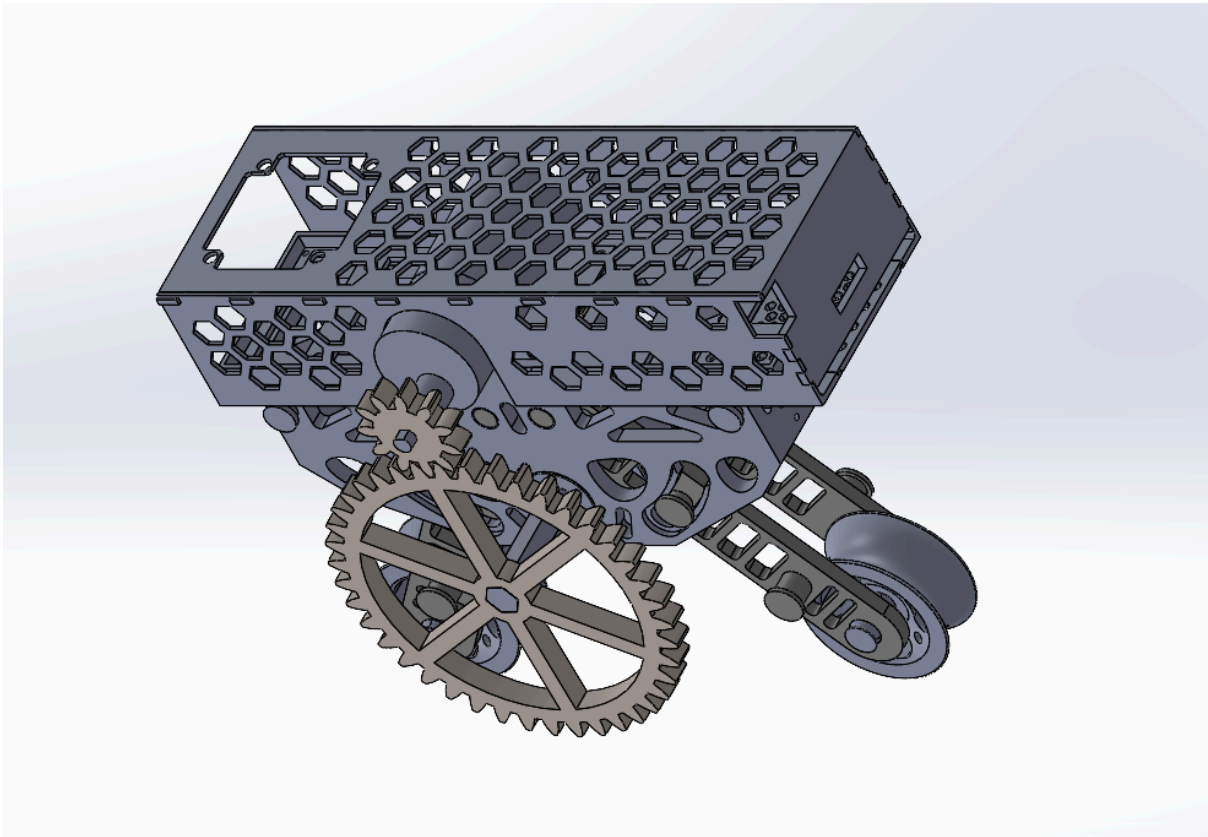
The output torque of the motorized wheel is given:

$$T_{wheel} = T_{motor} \times GR \times \eta$$

As the reduction gear ratio is 1:3 and the efficiency of 3d printed spur gears is taken to be 85% this gives us:

$$T_{wheel} = 59 \times 3 \times 0.85 = 150.45 \text{ Ncm}$$

giving a safety torque margin of $(150.45/74.94)$ **2.01**



Legs:

Starting with the legs, which we have 4 of, each of them is 5mm thick. We have used elongated slot pockets, perpendicular to the leg's length.

The key insight here is when a beam bends, the bending stress is the bending moment multiplied by the distance from the neutral axis divided by the moment of inertia. The material close to the neutral axis which is the middle region, contributes comparatively less to the bending stiffness. The walls between each slot ensures stress flow around the cutouts. We filleted edges to mitigate crack initiation sites. **The wall argument unsolved**

Body:

Organic slot cutouts, which are responsible for the reduction of 30-40% of the face area. The face panels show low displacement throughout, which tells us they weren't carrying significant load to begin with. Most of the structural stresses are taken by the edge walls and X-truss geometry. Now the X-truss geometry is used around the corners. There is a shear force generated from the motor torque and the weight of the heavy components. The triangulated corners resist the shear force by converting it into axial forces along the truss members which PLA plastic is great at handling.

Wheels:

The wheels follow the same philosophy, a 3 curved spoke-style cutouts between the hub and the outer rim. This removed material from the low-stress middle region whilst maintaining a strong structure between the hub of the wheel to the rim.

Conclusion:

The designs were validated in SolidWorks FEA, we applied exaggerated reaction forces from our calculations and ran a static displacement analysis. The maximum displacements are within acceptable limits for PLA under these load conditions {state max deflections for respective part}.

The optimisations weren't just aesthetic but helped reduce chassis mass by roughly 30 percent compared to a solid print, which feeds directly back into a lower torque requirement and a faster robot.

Fix the leg section and AI continuity check